

## CHE 396

### 1. Course number and name

CHE 396: Chemical Engineering Design I

### 2. Credits and contact hours

Credit hours: 4. Contact hours: Two, 70 min lecture periods; Engineering.

### 3. Instructor's or course coordinator's name

Betul Bilgin

### 4. Textbook, title, author, publisher, year

Towler, G., Sinnott, R., Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design, Second Edition, Elsevier, 2012.

#### a. Other supplemental materials

Seider, Seader, Lewin, Widagdo, Product and Process Design Principles, 3rd Edition, Wiley, 2009

Writing Style and Standards in Undergraduate Reports, 3e, by Jeffrey Donnell, Sheldon Jeter, Colin MacDougall, and Jacqueline Snedeker

### 5. Specific course information

#### a. Brief description of the content of the course (catalog description)

Introduction to modern, process design and development, engineering economics, and report writing. Design and cost of equipment relating to materials handling to heat transfer, mass transfer, and reactors.

#### b. Prerequisites or co-requisites

CHE 312, CHE 313, CHE 321

#### c. Indicate whether a required, elective, or selected elective.

Required

### 6. Specific goals for the course

#### a. Specific outcomes of instruction.

Provide a real-world perspective on process design

Introduce design and selection of equipment

Give insight into practical constraints that govern design decisions

Help you translate knowledge into practice & start solving real problem

Improve presentation, analytical thinking and team working skills

b. Explicitly indicate which of the student outcomes are addressed by the course.

| Student Learning Outcome                                                                                                                                                                                                                        | 1 | 2 | 3 | 4 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|
| (1) an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.                                                                                                  |   |   |   | X |
| (2) an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.                   |   |   |   | X |
| (3) an ability to communicate effectively with a range of audiences.                                                                                                                                                                            |   |   | X |   |
| (4) an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts. |   |   |   | X |
| (5) an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.                                                 |   |   |   | X |
| (6) an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.                                                                                                |   |   | X |   |
| (7) an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.                                                                                                                                             |   |   | X |   |

## 7. Brief list of topics to be covered

1. Chemical Design Introduction
2. Flowsheet development
3. Utilities and Energy Recovery
4. Process simulation
5. Process control
6. Materials of construction
7. Capital cost
8. Operating cost
9. Economic Analysis
10. Process Hazard, Professional ethics
11. Professional Ethics Presentations
12. General Site Consideration